

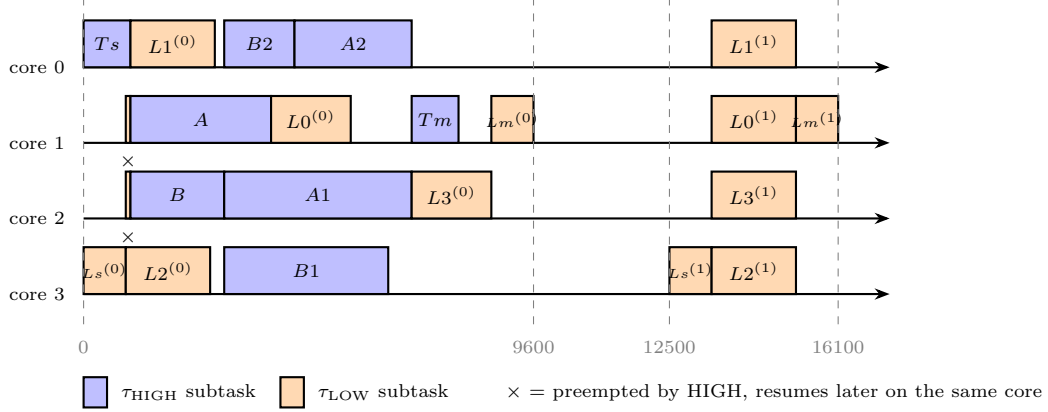


Figure 1: Simulated 4-core schedule under TDTA at 80Hz (LOW period = $12500\mu s$), respecting each task's DAG dependencies and SCHED_FIFO priority preemption (τ_{HIGH} over τ_{LOW}). Superscripts denote the LOW job index (job 0 released at $t=0$, job 1 at $t=12500$). $L0^{(0)}$ and $L3^{(0)}$ are each preempted once (marked $\times$) when τ_{HIGH} 's A/B become ready on the same core, then resume once that core frees up. Job 0's response time ($9600\mu s$) is far larger than job 1's ($3600\mu s$, $16100-12500$) purely from this phase difference between the two tasks' periods – the source of the run-to-run variance discussed in the text.